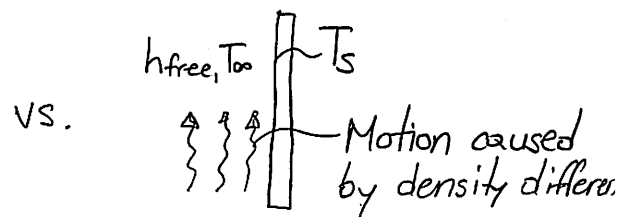
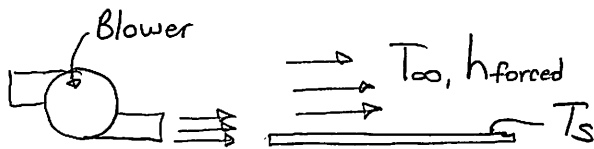


Convection Heat Transfer

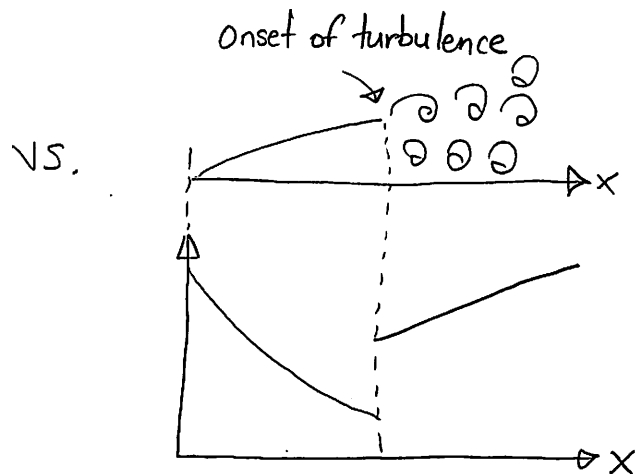
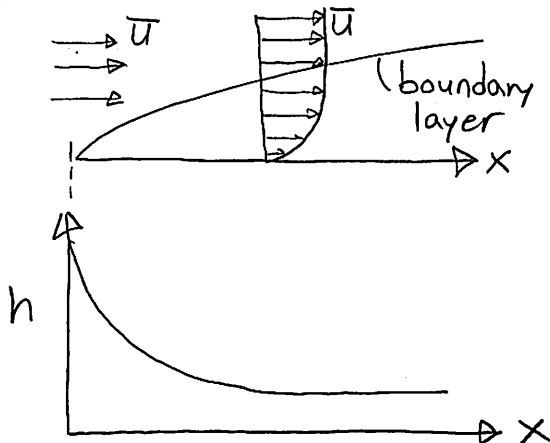
Convection is the transfer of thermal energy due to both conduction and by bulk "carrying" of the energy through the velocity of the fluid.

Four Categories

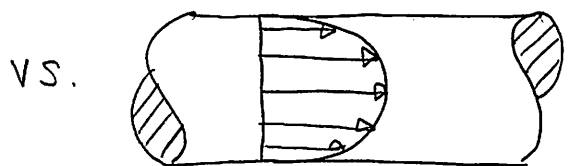
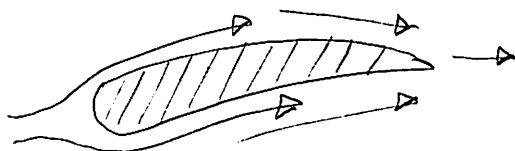
① Forced vs. Free (Natural)



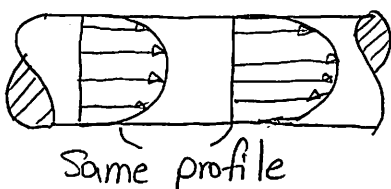
② Laminar vs. Turbulent



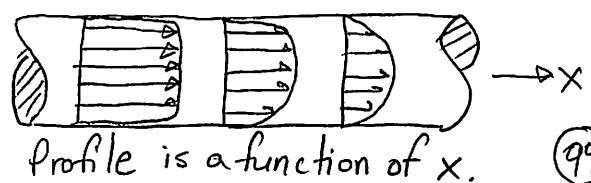
③ External vs. Internal



④ Fully Developed vs. Developing



VS



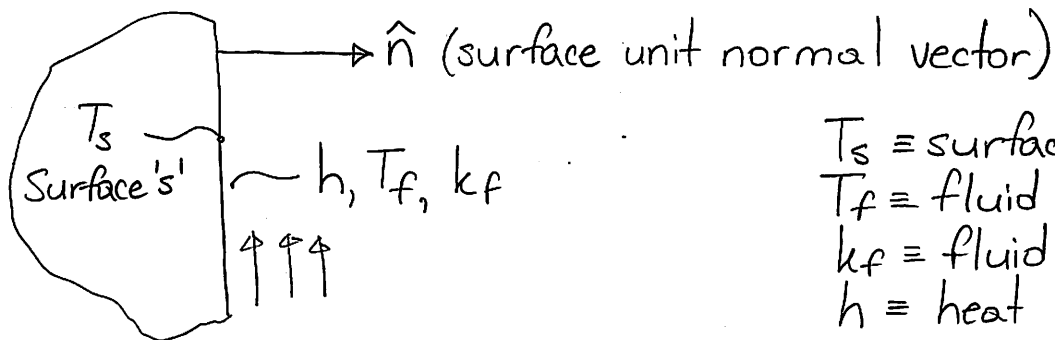
The Assumptions

- 1) Steady-State
- 2) Constant properties except where density differences cause the flow (Boussinesq approximation)
- 3) Incompressible

The Equations

- 1) Conservation of mass
- 2) Conservation of momentum (Navier-Stokes)
- 3) Conservation of energy (Heat equation for a fluid flow)

We've already seen convection before :



$T_s \equiv$ surface temperature
 $T_f \equiv$ fluid temperature
 $k_f \equiv$ fluid thermal cond.
 $h \equiv$ heat transfer coeff.

We know that :

$$q'' = h(T_s - T_f) \quad (1) \quad (\text{heat flux})$$

We can also write that :

$$q'' = -k_f \left. \frac{\partial T_f}{\partial \hat{n}} \right|_s \quad (2) \quad (\text{heat flux by Fourier's Law in fluid})$$

Let's non dimensionalize our problem :

$$\Theta = \frac{T - T_f}{T_s - T_f} ; \quad n^* = \frac{\hat{n}}{L}$$

Back substituting into ① & ② & equating the two:

$$-k_f \frac{T_s - T_f}{L} \cdot \frac{\partial \theta}{\partial n^*} \Big|_s = h (T_s - T_f)$$

$$\boxed{-\frac{\partial \theta}{\partial n^*} \Big|_s = \frac{hL}{k_f} = Nu} \Rightarrow \text{Nusselt Number}$$

$$\boxed{Nu = \frac{hL}{k_f} = \frac{\text{convective heat transfer rate in fluid}}{\text{conductive heat transfer rate in fluid}}}$$

↳ Note, very similar to Biot number (Bi) but

$$Bi = \frac{hL}{k_s} \leftarrow \text{solid thermal conductivity}$$

So Nu is a measure of the heat transfer enhancement you have due to bulk fluid motion when compared to conduction ^{only}.

Note, if $Nu = 1$, no convection effects & conduction only.

$$Nu = -\frac{\partial \theta}{\partial n^*} \Big|_s = 1 \Rightarrow \underbrace{-k_f \frac{T_s - T_f}{L}}_{\text{Conduction only}} = \underbrace{h (T_s - T_f)}_{\text{convection}}$$

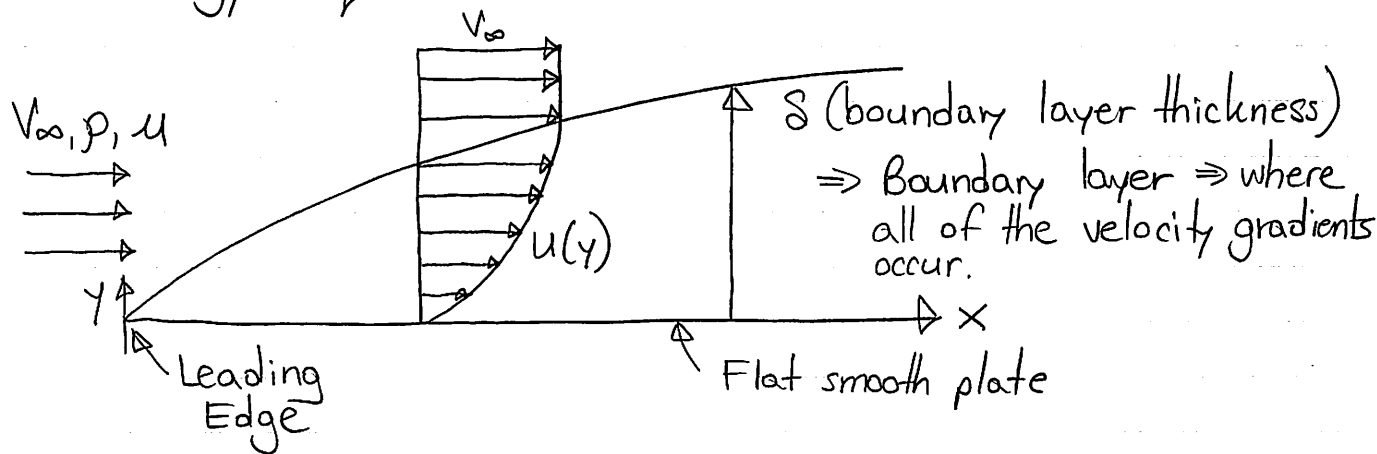
Since $Nu \sim \frac{\partial \theta}{\partial n^*} \Big|_s \sim \theta$, it typically depends on:

- 1) Flow conditions (laminar, turbulent)
- 2) Fluid properties
- 3) Geometry
- 4) Boundary conditions

External Flow

Assuming: 1) $\rho, \mu, C \equiv \text{constant}$
 2) Incompressible

We need to solve the coupled fluid flow (momentum) and energy equations to solve for h .



For a 2D laminar flow, flat plate, & steady

X-momentum:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right]$$

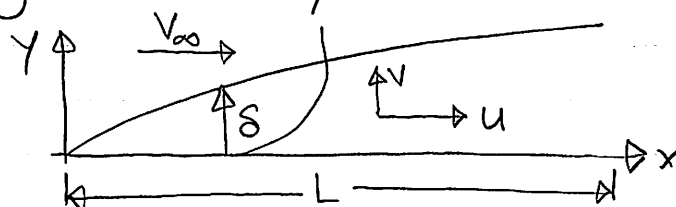
y-momentum:

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right]$$

continuity:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Looking more closely at our boundary layer:



Let's non-dimensionalize

Let: $\bar{u} = \frac{u}{V_\infty}$, $\bar{v} = \frac{v}{V_s}$, $\bar{p} = \frac{p}{\rho V_\infty^2}$

$\bar{x} = \frac{x}{L}$, $\bar{y} = \frac{y}{\delta}$

We know all of these parameters except for V_s .
Using continuity: $\bar{u} \sim V_\infty$, $\bar{x} \sim L$, $\bar{y} \sim \delta$ (" $\sim$ " means scales as)

$$\frac{\partial \bar{u}}{\partial \bar{x}} + \frac{\partial \bar{v}}{\partial \bar{y}} = 0 \Rightarrow \frac{V_\infty}{L} + \frac{V_s}{\delta} = 0 \Rightarrow \boxed{V_s \sim \left(\frac{\delta}{L}\right) V_\infty}$$

Note, when I use " $\sim$ ", it means scales as or is on the same order of magnitude as. Very powerfull method!

We know for boundary layers that $\delta \ll L$, hence $V_s \ll V_\infty$

So if we back substitute into our x-momentum

$$\begin{aligned} V_\infty \bar{u} \frac{V_\infty}{L} \frac{\partial \bar{u}}{\partial \bar{x}} + \left(\frac{\delta}{L}\right) V_\infty \bar{v} \frac{V_\infty}{\delta} \frac{\partial \bar{u}}{\partial \bar{y}} &= -\frac{1}{\rho} \cancel{\rho} \frac{V_\infty^2}{L} \frac{\partial \bar{p}}{\partial \bar{x}} + \nu \left[\frac{V_\infty}{L^2} \frac{\partial^2 \bar{u}}{\partial \bar{x}^2} + \frac{V_\infty}{\delta^2} \frac{\partial^2 \bar{u}}{\partial \bar{y}^2} \right] \\ \frac{V_\infty^2}{L} \bar{u} \frac{\partial \bar{u}}{\partial \bar{x}} + \frac{V_\infty^2}{L} \bar{v} \frac{\partial \bar{u}}{\partial \bar{y}} &= -\frac{V_\infty^2}{L} \frac{\partial \bar{p}}{\partial \bar{x}} + \frac{V_\infty \nu}{L^2} \left[\frac{\partial^2 \bar{u}}{\partial \bar{x}^2} + \underbrace{\left(\frac{L^2}{\delta^2}\right)}_{\gg 1, \text{ so dominate}} \frac{\partial^2 \bar{u}}{\partial \bar{y}^2} \right] \end{aligned}$$

Divide both sides by $\frac{V_\infty^2}{L}$

$$\bar{u} \frac{\partial \bar{u}}{\partial \bar{x}} + \bar{v} \frac{\partial \bar{u}}{\partial \bar{y}} = -\frac{\partial \bar{p}}{\partial \bar{x}} + \underbrace{\frac{\nu}{V_\infty L}}_{\frac{1}{Re_L}} \frac{\partial^2 \bar{u}}{\partial \bar{y}^2}$$

Note, these are all ~ 1 .

$$\frac{1}{Re_L} \Rightarrow \boxed{Re_L = \frac{V_\infty L}{\nu} = \frac{\rho V_\infty L}{\mu}}$$

Where: $Re_L \equiv \text{Reynolds Number} \equiv \frac{\text{inertial force}}{\text{viscous force}}$

$$\boxed{\bar{u} \frac{\partial \bar{u}}{\partial \bar{x}} + \bar{v} \frac{\partial \bar{u}}{\partial \bar{y}} = -\frac{\partial \bar{p}}{\partial \bar{x}} + \frac{1}{Re_L} \frac{\partial^2 \bar{u}}{\partial \bar{y}^2}} \quad (1) \rightarrow \text{Non-dimensional x-momentum equation}$$

Let's now look at y-momentum:

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right] \Rightarrow \text{Non-dimensionalize this}$$

$$V_\infty \bar{u} \left(\frac{\delta}{L} \right) V_\infty \cdot \frac{1}{L} \frac{\partial \bar{v}}{\partial \bar{x}} + \left(\frac{\delta}{L} \right) V_\infty \bar{v} \left(\frac{\delta}{L} \right) V_\infty \frac{1}{\delta} \frac{\partial \bar{v}}{\partial \bar{y}} = -\frac{\partial \bar{p}}{\partial \bar{y}} + \frac{\nu V_\infty^2}{\delta^3} \left[\left(\frac{\delta}{L} \right) V_\infty \frac{1}{L^2} \frac{\partial^2 \bar{v}}{\partial \bar{x}^2} + \left(\frac{\delta}{L} \right) V_\infty \frac{1}{\delta^2} \frac{\partial^2 \bar{v}}{\partial \bar{y}^2} \right]$$

Collecting all terms and multiplying both sides by δ

$$V_\infty^2 \left(\frac{\delta^2}{L^2} \right) \bar{u} \frac{\partial \bar{v}}{\partial \bar{x}} + V_\infty^2 \left(\frac{\delta^2}{L^2} \right) \bar{v} \frac{\partial \bar{v}}{\partial \bar{y}} = -V_\infty^2 \frac{\partial \bar{p}}{\partial \bar{y}} + \nu V_\infty \left[\left(\frac{\delta^2}{L^3} \right) \frac{\partial^2 \bar{v}}{\partial \bar{x}^2} + \frac{1}{L} \frac{\partial^2 \bar{v}}{\partial \bar{y}^2} \right]$$

Divide both sides by V_∞^2 :

$$\left(\frac{\delta^2}{L^2} \right) \bar{u} \frac{\partial \bar{v}}{\partial \bar{x}} + \left(\frac{\delta^2}{L^2} \right) \bar{v} \frac{\partial \bar{v}}{\partial \bar{y}} = -\frac{\partial \bar{p}}{\partial \bar{y}} + \frac{\nu}{V_\infty L} \left[\left(\frac{\delta^2}{L^2} \right) \frac{\partial^2 \bar{v}}{\partial \bar{x}^2} + \frac{\partial^2 \bar{v}}{\partial \bar{y}^2} \right]$$

Since $\left(\frac{\delta}{L} \right)^2 \ll 1$ (Boundary Layer approx), many of our terms drop out.

$$\boxed{\frac{\partial \bar{p}}{\partial \bar{y}} = \frac{1}{Re_L} \frac{\partial^2 \bar{v}}{\partial \bar{y}^2}} \quad (2) \rightarrow \text{Non-dimensional y-momentum}$$

It turns out you can use scaling analysis to show that

$$\frac{\partial^2 \bar{v}}{\partial \bar{y}^2} \ll \frac{\partial^2 \bar{u}}{\partial \bar{y}^2} \Rightarrow \frac{\partial \bar{p}}{\partial \bar{y}} \approx 0 \Rightarrow \boxed{\rho = f(x) \text{ only}}$$

↳ Derived in ME521 (Convective Heat Transfer)